



Real-Time Quality Control: How HPCL Uses Industrial AI to Improve Refining Processes

15-20%

Cost Reductions in
Sampling & Analysis

Up to \$400K

Savings in Product
Quality

CHALLENGE

HPCL's Mumbai Refinery faced a major issue with product quality giveaways, as assessing product quality was both challenging and time-consuming.

SOLUTION

The refining team used Aspen AI Model Builder™ and Aspen HYSYS® to build and deploy AI-based soft sensors that could predict the desired product quality parameters in real-time.

VALUE CREATED

- Real-time visibility into product quality has enabled operators to take corrective actions that have helped the refinery save an estimated \$100,000-\$400,000 USD per year.
- Real-time quality insights have also reduced the need for frequent sampling and analysis, cutting those costs by 15-20%.





Challenge

Hindustan Petroleum Corporation Limited (HPCL) is a refining company in India, owned by the Oil and Natural Gas Corporation (ONGC), which is a government entity. HPCL owns two refineries: Mumbai Refinery (capacity of 7.5 MMTPA) and Visakh Refinery (capacity of 8.3 MMTPA).

HPCL's Mumbai Refinery faced significant challenges with high operational expenditures and low margins, which affected its profitability. Frequent crude changeovers disrupted the optimal operating conditions of various process units, thereby impacting the refinery's operational efficiency.

A major issue was product quality giveaways. Like many refineries, assessing product quality was both challenging and time-consuming, involving manual collection of liquid samples from the plant and delays due to the subsequent chemical laboratory analysis. These delays resulted in lost opportunities for process optimization. Additionally, the manual collection of liquid samples increased the risk of sampling errors.



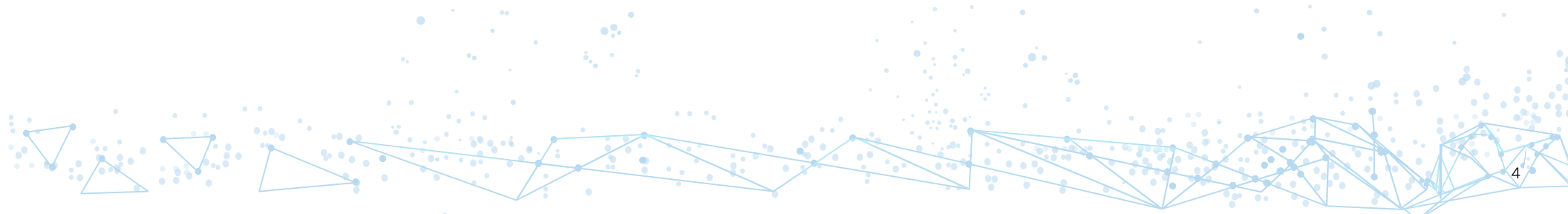


Solution

The refining team addressed this issue by building and deploying **AI-based soft sensors** that predict the desired product quality parameters in real-time every three minutes. The soft sensors were built using Aspen AI Model Builder (AIMB) and Aspen HYSYS. Soft sensors are a collection of models that can be used to accurately predict and forecast infrequently measured product properties (such as laboratory measurements of petroleum products) based on more frequent measurements of product parameters like temperature, pressure and flow rate provided by physical sensors.

The team at HPCL Mumbai Refinery created a soft sensor model to predict the viscosity of the vacuum gas oil stream, which is the first side stream from the vacuum distillation unit. This real-time model, developed using AIMB and Aspen HYSYS, combines the benefits of conventional rigorous process simulation with the power of AI.

AIMB helped the team build and validate accurate AI models. It allowed them to identify factors affecting viscosity measurements and provided various data cleaning options. The team developed multiple models using different machine learning methods and transformations, then tested these models to find the most accurate one.



Additionally, AIMB enabled rigorous validation of the model's accuracy against validation data. The model predictions were highly accurate, as shown in Figure 1. (The two mismatched points in the graph occurred during abnormal plant operations, but the models performed very well during normal operations.)

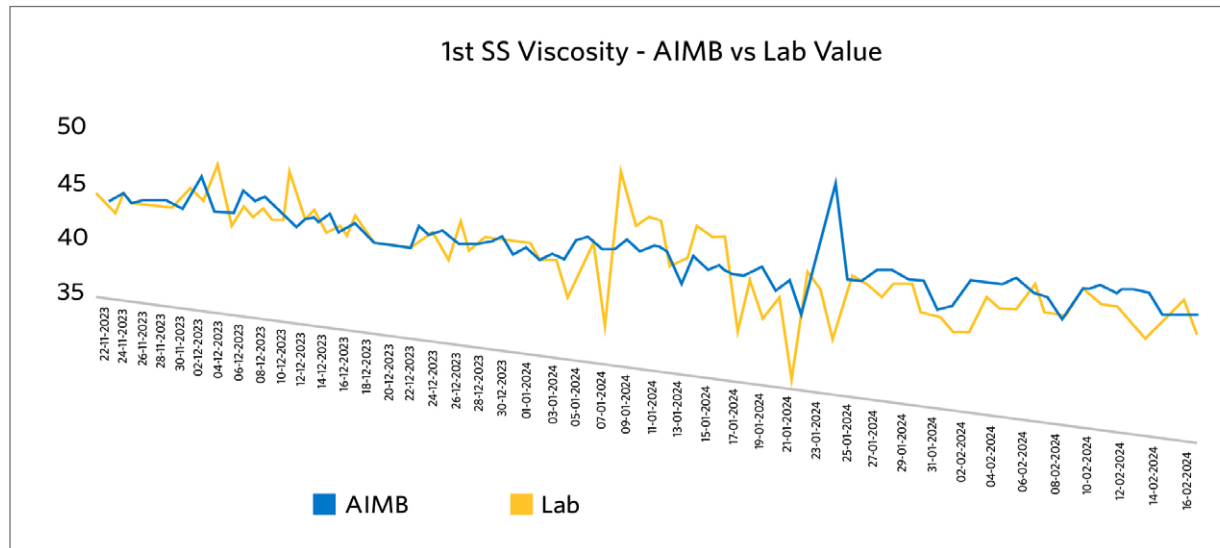


Figure 1. The model predictions of viscosity values were shown to be accurate.



Value Created

The soft sensor model built with AIMB was deployed in Aspen HYSYS. It runs automatically every three minutes, calibrating itself to real-time plant data and calculating viscosity values. This gives the team real-time visibility into product quality, allowing operators to optimize the process without waiting for lab results. Prompt corrective actions have helped the refinery minimize quality giveaways and maximize yield, saving an estimated \$100,000-\$400,000 USD per year.

Additionally, real-time quality insights have reduced the need for frequent sampling and reliance on lab analysis, cutting associated costs by around 15-20%. The refinery expects to further reduce lab analysis and save costs as they gain more confidence in the model's accuracy.

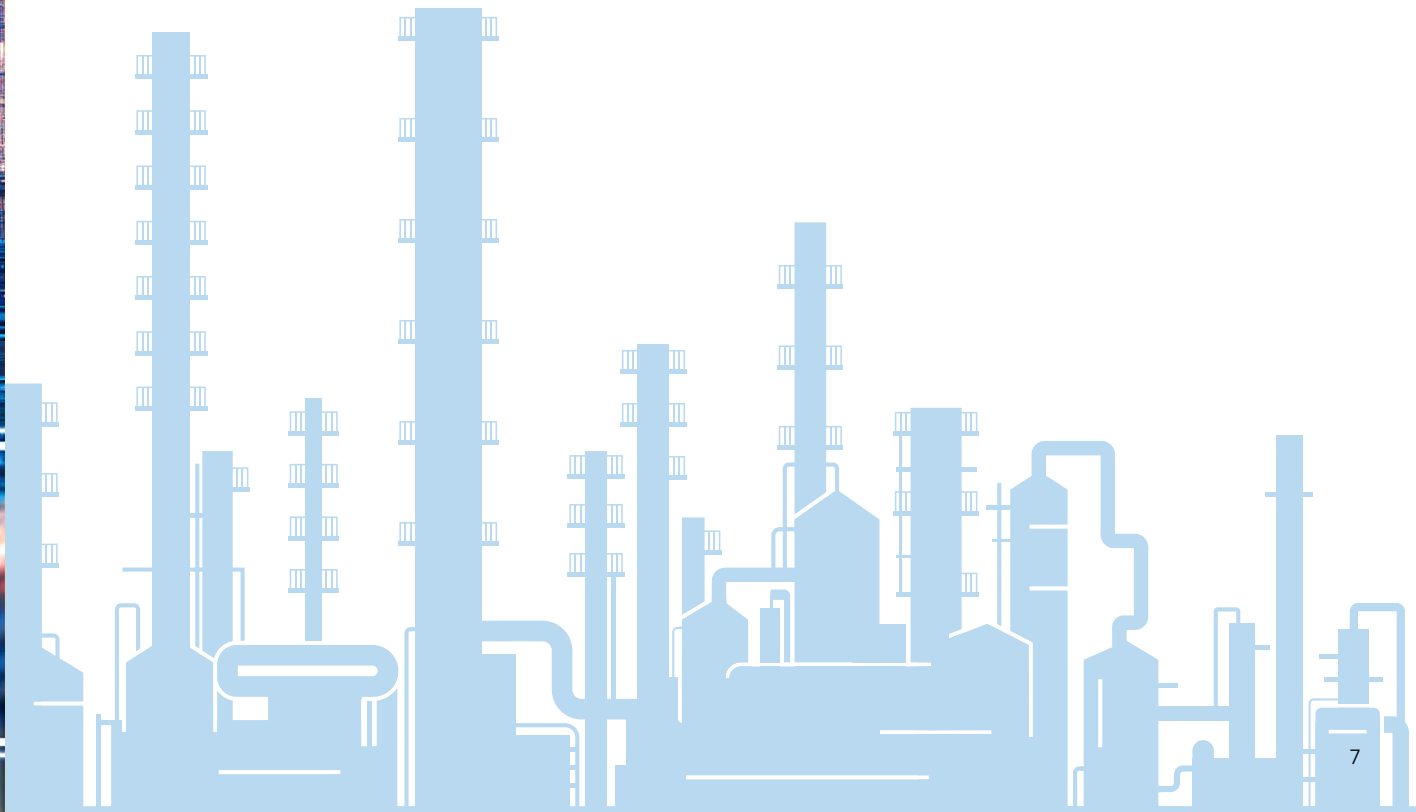




Conclusion

HPCL Mumbai Refinery's implementation of AI-based soft sensors has significantly enhanced their operational efficiency and profitability. By predicting product quality parameters in real-time, the refinery has minimized quality giveaways and maximized yield, resulting in substantial annual savings. The deployment of these models has also reduced the need for frequent sampling and reliance on laboratory analysis, further cutting costs.

As the team continues to gain confidence in the accuracy of these AI models, they anticipate even greater reductions in operational expenditures and improved process optimization. This innovative approach demonstrates the transformative potential of AI in refining operations.





About Aspen Technology

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